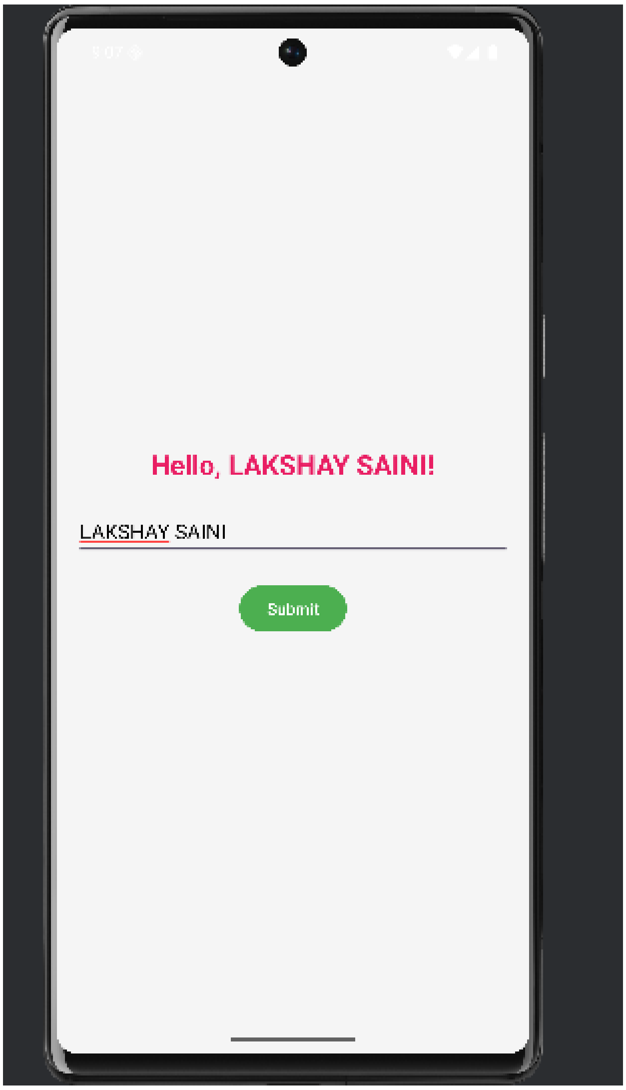
1. Write a program to develop an application that uses GUI components, Font and Colors.

<u>Activity_main.xml:-</u>

```xml
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
  android:orientation="vertical"
  android:gravity="center"
  android:background="#F5F5F5"
  android:padding="16dp"
  android:layout_width="match_parent"
  android:layout_height="match_parent">

  <TextView
    android:id="@+id/titleText"
    android:text="Welcome to Android GUI"
    android:textSize="24sp"
    android:textStyle="bold"
    android:textColor="#2196F3"
    android:layout_marginBottom="20dp"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content" />

  <EditText
    android:id="@+id/editName"
    android:hint="Enter your name"
    android:textColor="#000000"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"/>

  <Button
    android:id="@+id/btnSubmit"
    android:text="Submit"
    android:backgroundTint="#4CAF50"
    android:textColor="#FFFFFF"
    android:layout_marginTop="20dp"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"/>

</LinearLayout>
```

Mainactivity.java:-

```
package com.example.androidpracticals;
import androidx.appcompat.app.AppCompatActivity;
import android.os.Bundle;
import android.widget.*;

public class MainActivity extends AppCompatActivity {
  @Override
  protected void onCreate(Bundle savedInstanceState) {
    super.onCreate(savedInstanceState);
    setContentView(R.layout.activity_main);

    EditText editName = findViewById(R.id.editName);
    Button btnSubmit = findViewById(R.id.btnSubmit);
    TextView titleText = findViewById(R.id.titleText);

    btnSubmit.setOnClickListener(v -> {
      String name = editName.getText().toString();
      titleText.setText("Hello, " + name + "!");
      titleText.setTextColor(0xFFE91E63);
    });
  }
}
```


Hello, LAKSHAY SAINI!
LAKSHAY SAINI
Submit

2. Write a program to develop an application that uses Layout Managers and Event Listeners.

<u>Activity_main.xml:-</u>

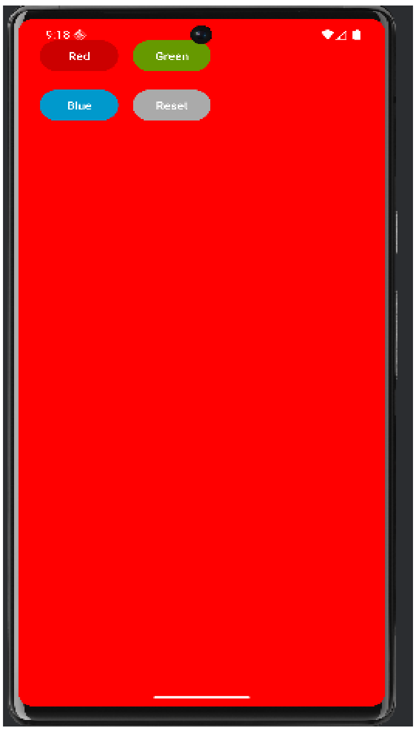
```
<?xml version="1.0" encoding="utf-8"?>
<GridLayout xmlns:android="http://schemas.android.com/apk/res/android"
  android:id="@+id/gridLayout"
  android:layout_width="match_parent"
  android:layout_height="match_parent"
  android:columnCount="2"
  android:rowCount="2"
  android:orientation="horizontal"
  android:padding="16dp">

  <Button
    android:id="@+id/btnRed"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="Red"
    android:backgroundTint="@android:color/holo_red_dark"
    android:textColor="@android:color/white"
    android:layout_margin="8dp" />

  <Button
    android:id="@+id/btnGreen"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="Green"
    android:backgroundTint="@android:color/holo_green_dark"
    android:textColor="@android:color/white"
    android:layout_margin="8dp" />

  <Button
    android:id="@+id/btnBlue"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="Blue"
    android:backgroundTint="@android:color/holo_blue_dark"
    android:textColor="@android:color/white"
```

```
    android:layout_margin="8dp" />

  <Button
    android:id="@+id/btnReset"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="Reset"
    android:backgroundTint="@android:color/darker_gray"
    android:textColor="@android:color/white"
    android:layout_margin="8dp" />

</GridLayout>
```

<u>Mainactivity.java:-</u>

```java
package com.example.androidpracticals;

import android.graphics.Color;
import android.os.Bundle;
import android.widget.Button;
import android.widget.GridLayout;
import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {

  private GridLayout layout;
  private Button btnRed, btnGreen, btnBlue, btnReset;

  @Override
  protected void onCreate(Bundle savedInstanceState) {
    super.onCreate(savedInstanceState);
    setContentView(R.layout.activity_main);

    initializeViews();
    setupListeners();
  }

  /** Initialize all UI components */
  private void initializeViews() {
    layout = findViewById(R.id.gridLayout);
```

```
        btnRed = findViewById(R.id.btnRed);
        btnGreen = findViewById(R.id.btnGreen);
        btnBlue = findViewById(R.id.btnBlue);
        btnReset = findViewById(R.id.btnReset);
    }

    /** Set up button click listeners */
    private void setupListeners() {
        btnRed.setOnClickListener(v -> changeBackgroundColor(Color.RED));
        btnGreen.setOnClickListener(v -> changeBackgroundColor(Color.GREEN));
        btnBlue.setOnClickListener(v -> changeBackgroundColor(Color.BLUE));
        btnReset.setOnClickListener(v -> changeBackgroundColor(Color.WHITE));
    }

    /** Helper method to change background color */
    private void changeBackgroundColor(int color) {
        layout.setBackgroundColor(color);
    }
}
```

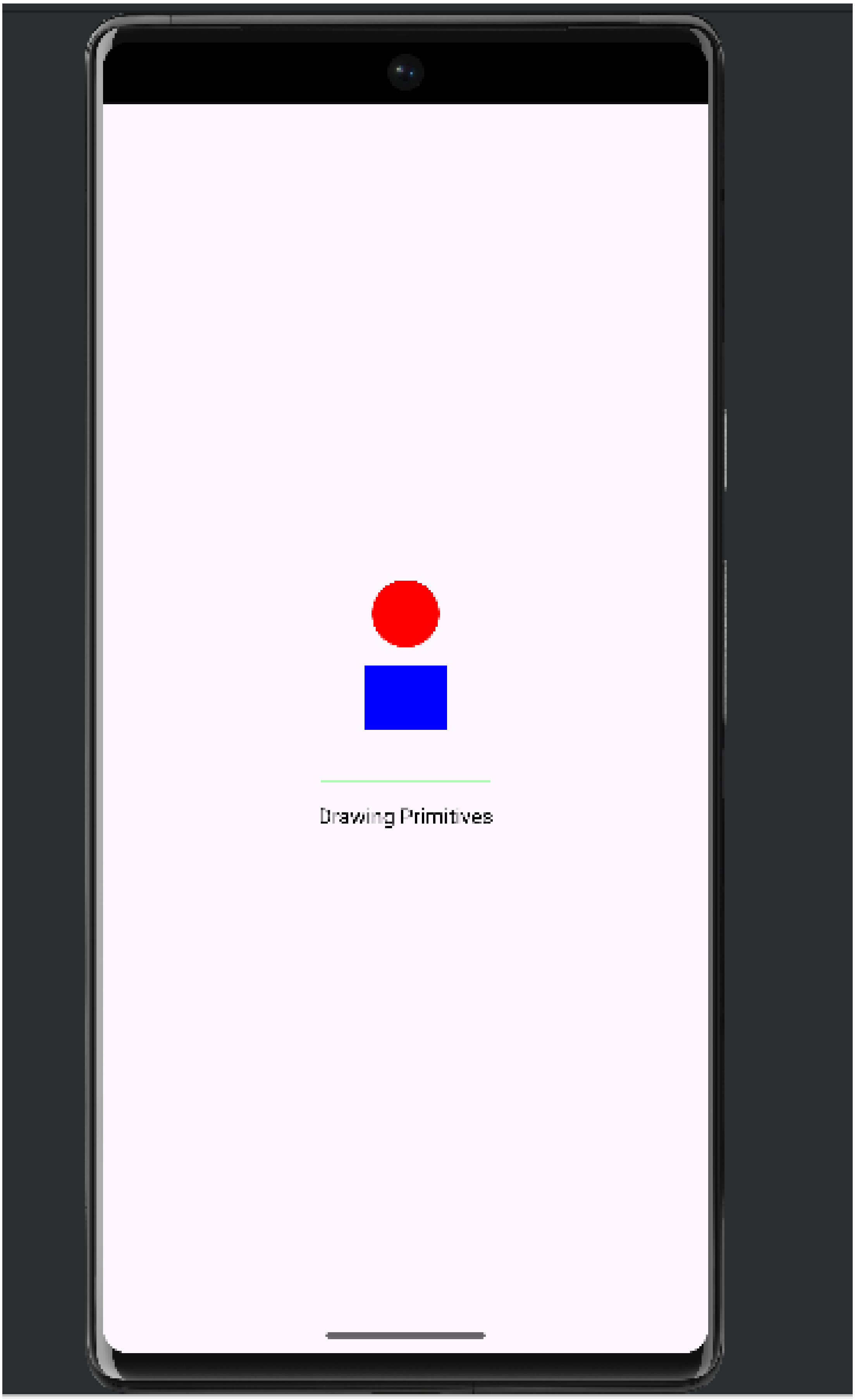


3. Write a program for an application that draws basic Graphical Primitives on the screen.

<u>Mainactivity.java:-</u>

```
package com.example.androidpracticals;

import androidx.appcompat.app.AppCompatActivity;
import android.os.Bundle;
import android.content.Context;
import android.graphics.*;
import android.view.View;

public class MainActivity extends AppCompatActivity {

  @Override
  protected void onCreate(Bundle savedInstanceState) {
    super.onCreate(savedInstanceState);
    setContentView(new MyCanvas(this));
  }

  class MyCanvas extends View {
    Paint paint = new Paint();

    MyCanvas(Context c) {
      super(c);
    }

    @Override
    protected void onDraw(Canvas canvas) {
      super.onDraw(canvas);

      // Get screen width and height dynamically
      int width = getWidth();
      int height = getHeight();

      // Calculate center point
      float cx = width / 2f;
      float cy = height / 2f;
```

```
        // Draw red circle at center
        paint.setColor(Color.RED);
        canvas.drawCircle(cx, cy - 200, 80, paint);

        // Draw blue rectangle below circle
        paint.setColor(Color.BLUE);
        float rectWidth = 200;
        float rectHeight = 150;
        canvas.drawRect(cx - rectWidth / 2, cy - rectHeight / 2, cx + rectWidth / 2,
cy + rectHeight / 2, paint);

        // Draw green line below rectangle
        paint.setColor(Color.GREEN);
        canvas.drawLine(cx - 200, cy + 200, cx + 200, cy + 200, paint);

        // Draw black text below line
        paint.setColor(Color.BLACK);
        paint.setTextSize(50);
        paint.setTextAlign(Paint.Align.CENTER);
        canvas.drawText("Drawing Primitives", cx, cy + 300, paint);
      }
   }
}
```


Drawing Primitives

4. Write a program to develop an application that make use of databases.

<u>Activity_main.xml:-</u>

```
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
  android:orientation="vertical"
  android:padding="16dp"
  android:gravity="center_horizontal"
  android:layout_width="match_parent"
  android:layout_height="match_parent">

  <EditText
    android:id="@+id/etName"
    android:hint="Enter Name"
    android:layout_width="match_parent"
    android:layout_height="wrap_content" />

  <EditText
    android:id="@+id/etEmail"
    android:hint="Enter Email"
    android:layout_width="match_parent"
    android:layout_height="wrap_content" />

  <Button
    android:id="@+id/btnAdd"
    android:text="Add Data"
    android:layout_width="match_parent"
    android:layout_height="wrap_content" />

  <Button
    android:id="@+id/btnView"
    android:text="View Data"
    android:layout_width="match_parent"
    android:layout_height="wrap_content" />

  <TextView
    android:id="@+id/tvResult"
```

```
    android:text="Results will appear here"
    android:padding="10dp"
    android:textSize="16sp"
    android:layout_width="match_parent"
    android:layout_height="wrap_content" />
</LinearLayout>
```

DatabaseHelper.java:-

```java
package com.example.androidpracticals;

import android.content.ContentValues;
import android.content.Context;
import android.database.Cursor;
import android.database.sqlite.SQLiteDatabase;
import android.database.sqlite.SQLiteOpenHelper;

public class DatabaseHelper extends SQLiteOpenHelper {

  public static final String DATABASE_NAME = "StudentDB.db";
  public static final String TABLE_NAME = "student_table";
  public static final String COL_1 = "ID";
  public static final String COL_2 = "NAME";
  public static final String COL_3 = "EMAIL";

  public DatabaseHelper(Context context) {
    super(context, DATABASE_NAME, null, 1);
  }

  @Override
  public void onCreate(SQLiteDatabase db) {
    db.execSQL("CREATE TABLE " + TABLE_NAME +
        " (ID INTEGER PRIMARY KEY AUTOINCREMENT, NAME TEXT, EMAIL TEXT)");
  }

  @Override
  public void onUpgrade(SQLiteDatabase db, int oldVersion, int newVersion) {
```

```
        db.execSQL("DROP TABLE IF EXISTS " + TABLE_NAME);
        onCreate(db);
    }

    public boolean insertData(String name, String email) {
        SQLiteDatabase db = this.getWritableDatabase();
        ContentValues contentValues = new ContentValues();
        contentValues.put(COL_2, name);
        contentValues.put(COL_3, email);
        long result = db.insert(TABLE_NAME, null, contentValues);
        return result != -1;
    }

    public Cursor getAllData() {
        SQLiteDatabase db = this.getWritableDatabase();
        return db.rawQuery("SELECT * FROM " + TABLE_NAME, null);
    }
}
```

MainActivity.java:-

```
package com.example.androidpracticals;

import androidx.appcompat.app.AppCompatActivity;
import android.database.Cursor;
import android.os.Bundle;
import android.view.View;
import android.widget.*;

public class MainActivity extends AppCompatActivity {

    EditText etName, etEmail;
    Button btnAdd, btnView;
    TextView tvResult;
    DatabaseHelper db;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
```

```
        setContentView(R.layout.activity_main);

        db = new DatabaseHelper(this);

        etName = findViewById(R.id.etName);
        etEmail = findViewById(R.id.etEmail);
        btnAdd = findViewById(R.id.btnAdd);
        btnView = findViewById(R.id.btnView);
        tvResult = findViewById(R.id.tvResult);

        btnAdd.setOnClickListener(v -> {
            boolean inserted = db.insertData(etName.getText().toString(),
etEmail.getText().toString());
            if (inserted)
                Toast.makeText(this, "Data Inserted", Toast.LENGTH_SHORT).show();
            else
                Toast.makeText(this, "Insert Failed", Toast.LENGTH_SHORT).show();
        });

        btnView.setOnClickListener(v -> {
            Cursor res = db.getAllData();
            if (res.getCount() == 0) {
                tvResult.setText("No Data Found");
                return;
            }

            StringBuilder buffer = new StringBuilder();
            while (res.moveToNext()) {
                buffer.append("ID: ").append(res.getString(0)).append("\n");
                buffer.append("Name: ").append(res.getString(1)).append("\n");
                buffer.append("Email: ").append(res.getString(2)).append("\n\n");
            }
            tvResult.setText(buffer.toString());
        });
    }
}
```


Enter Name
Enter Email
Add Data
View Data
ID: 1
Name:
Email:
ID: 2
Name: LAKSHAY SAINI
Email: lakshay22saini@gmail.com
ID: 3
Name: Abhay Madhuwan
Email: adbay02madhuwan@gmail.com

5. Write a program to develop that make use of Notification Manager.

Activity_main.xml:-

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
   xmlns:tools="http://schemas.android.com/tools"
   android:layout_width="match_parent"
   android:layout_height="match_parent"
   android:orientation="vertical"
   android:gravity="center"
   android:background="#FFFFFF"
   tools:context=".MainActivity">

   <Button
      android:id="@+id/btnNotify"
      android:layout_width="wrap_content"
      android:layout_height="wrap_content"
      android:text="@string/show_notification"
      android:padding="12dp"
      android:backgroundTint="@color/purple_500"
      android:textColor="@android:color/white" />

</LinearLayout>
```

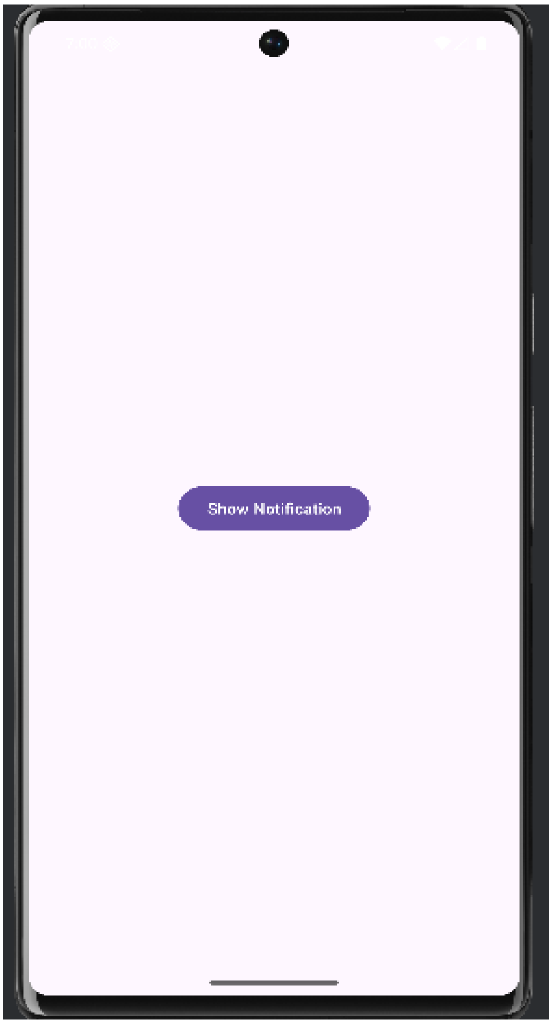
MainActivity.java:-

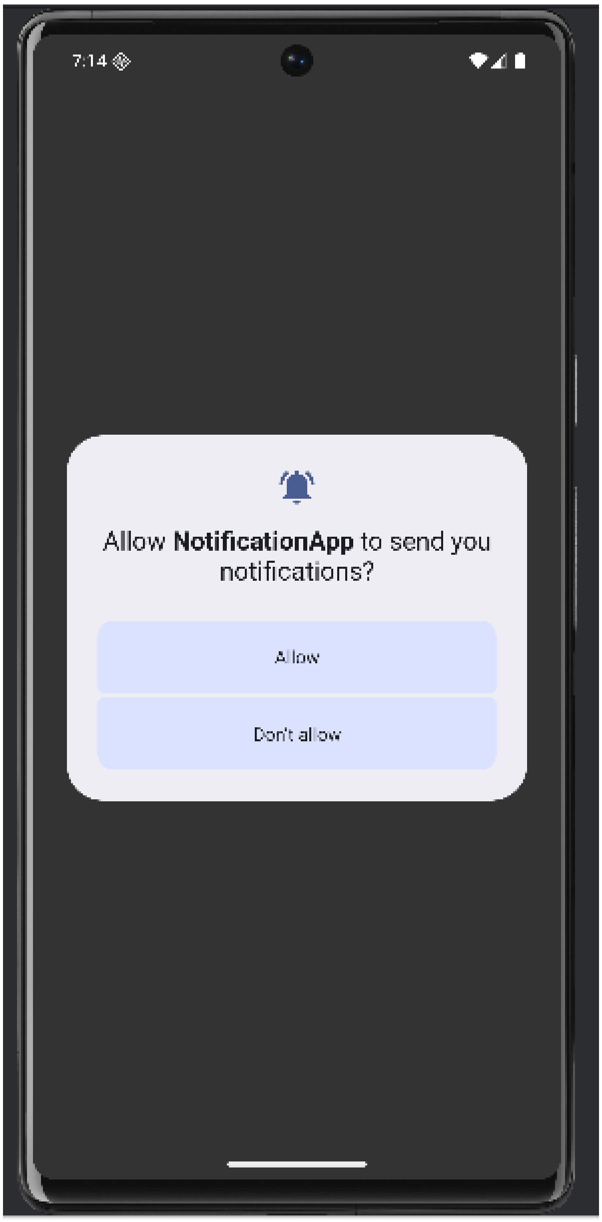
```
package com.example.androidpracticals;

import androidx.activity.result.ActivityResultLauncher;
import androidx.activity.result.contract.ActivityResultContracts;
//import androidx.annotation.NonNull;
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.app.NotificationCompat;
import androidx.core.app.NotificationManagerCompat;
import androidx.core.content.ContextCompat;

import android.Manifest;
```

```
import android.app.NotificationChannel;
import android.app.NotificationManager;
import android.app.PendingIntent;
import android.content.Intent;
import android.content.pm.PackageManager;
import android.os.Build;
import android.os.Bundle;
//import android.view.View;
import android.widget.Button;
import android.widget.Toast;

public class MainActivity extends AppCompatActivity {

    private static final String CHANNEL_ID = "channel_1";
    private static final int NOTIFICATION_ID = 1001;

    // Permission launcher for notifications
    private final ActivityResultLauncher<String> requestPermissionLauncher =
            registerForActivityResult(new ActivityResultContracts.RequestPermission(),
isGranted -> {
                if (isGranted) {
                    showNotification();
                } else {
                    Toast.makeText(this, "Notification permission denied",
Toast.LENGTH_SHORT).show();
                }
            });

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        Button btnNotify = findViewById(R.id.btnNotify);
        createNotificationChannel();

        btnNotify.setOnClickListener(view -> checkAndShowNotification());
    }

    private void checkAndShowNotification() {
        if (Build.VERSION.SDK_INT >= Build.VERSION_CODES.TIRAMISU) {
```

```
        if (ContextCompat.checkSelfPermission(this,
Manifest.permission.POST_NOTIFICATIONS)
            == PackageManager.PERMISSION_GRANTED) {
          showNotification();
        } else {
          // Request permission if not granted

requestPermissionLauncher.launch(Manifest.permission.POST_NOTIFICATIONS);
        }
      } else {
        showNotification();
      }
    }

    private void showNotification() {
      Intent intent = new Intent(this, MainActivity.class);
      intent.setFlags(Intent.FLAG_ACTIVITY_NEW_TASK |
Intent.FLAG_ACTIVITY_CLEAR_TASK);

      PendingIntent pendingIntent = PendingIntent.getActivity(
          this, 0, intent, PendingIntent.FLAG_IMMUTABLE
      );

      NotificationCompat.Builder builder = new NotificationCompat.Builder(this,
CHANNEL_ID)
          .setSmallIcon(R.mipmap.ic_launcher)
          .setContentTitle("Android Practical 5")
          .setContentText("This is your sample notification — tap to open the app!")
          .setPriority(NotificationCompat.PRIORITY_DEFAULT)
          .setContentIntent(pendingIntent)
          .setAutoCancel(true)
          .setDefaults(NotificationCompat.DEFAULT_ALL);

      NotificationManagerCompat manager = NotificationManagerCompat.from(this);

      if (ContextCompat.checkSelfPermission(this,
Manifest.permission.POST_NOTIFICATIONS)
          == PackageManager.PERMISSION_GRANTED) {
        manager.notify(NOTIFICATION_ID, builder.build());
      }
    }
```

```
    private void createNotificationChannel() {
        if (Build.VERSION.SDK_INT >= Build.VERSION_CODES.O) {
            CharSequence name = "Default Channel";
            String description = "Channel for Practical 5 Notifications";
            int importance = NotificationManager.IMPORTANCE_DEFAULT;

            NotificationChannel channel = new NotificationChannel(CHANNEL_ID, name, importance);
            channel.setDescription(description);

            NotificationManager manager = getSystemService(NotificationManager.class);
            if (manager != null) {
                manager.createNotificationChannel(channel);
            }
        }
    }
}
```




7:14
Allow NotificationApp to send you notifications?
Allow
Don't allow

6. Write a program to implement an application that uses Multi-threading.

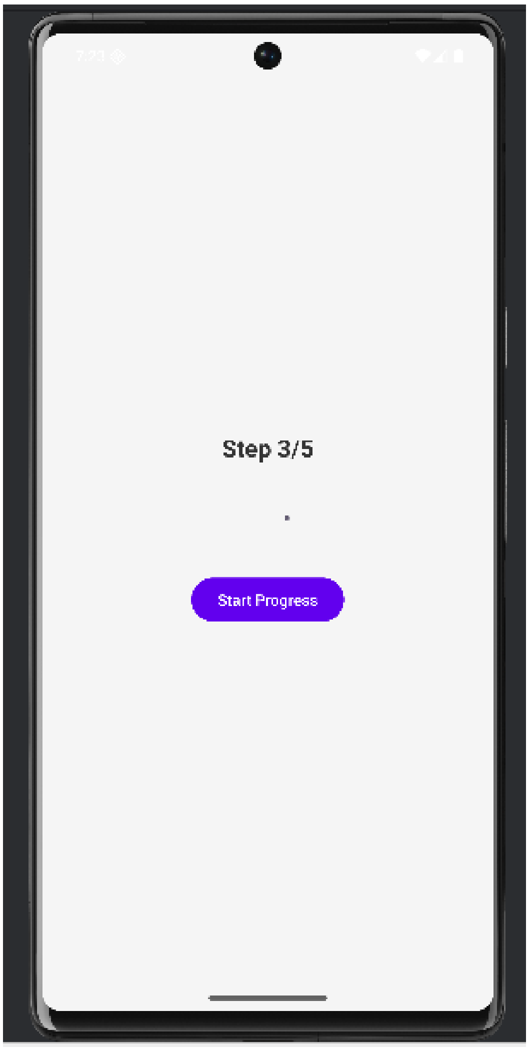
Activity_main.xml:-

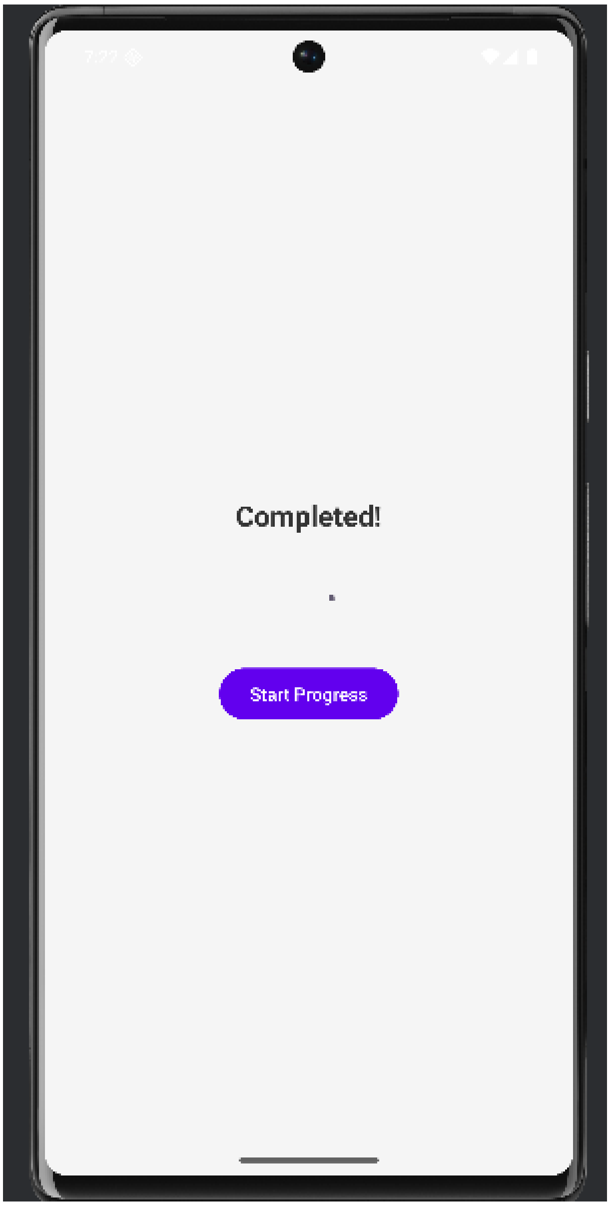
```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
  xmlns:android="http://schemas.android.com/apk/res/android"
  android:layout_width="match_parent"
  android:layout_height="match_parent"
  android:orientation="vertical"
  android:gravity="center"
  android:padding="24dp"
  android:background="#F5F5F5">

  <TextView
    android:id="@+id/textProgress"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="@string/progress_text"
    android:textSize="22sp"
    android:textStyle="bold"
    android:textColor="#333333"
    android:padding="12dp" />

  <ProgressBar
    android:id="@+id/progressBar"
    android:layout_width="250dp"
    android:layout_height="wrap_content"
    android:layout_marginTop="16dp"
    android:indeterminate="false"
    android:max="100"
    android:progress="0"
    android:progressTint="#6200EE" />

  <Button
    android:id="@+id/btnStart"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:layout_marginTop="24dp"
```

```
    android:text="@string/start_progress"
    android:textAllCaps="false"
    android:textColor="#FFFFFF"
    android:backgroundTint="#6200EE" />
</LinearLayout>
```

MainActivity.java:-

```
package com.example.androidpracticals;
import androidx.appcompat.app.AppCompatActivity;
import android.os.Bundle;
import android.widget.TextView;

public class MainActivity extends AppCompatActivity {
  TextView tv;
  @Override
  protected void onCreate(Bundle s) {
    super.onCreate(s);
    setContentView(R.layout.activity_main);
    tv = findViewById(R.id.textProgress);

    new Thread(() -> {
      for (int i=1;i<=5;i++){
        int finalI = i;
        runOnUiThread(() -> tv.setText("Step " + finalI + "/5"));
        try { Thread.sleep(1000); } catch (InterruptedException ignored) {}
      }
      runOnUiThread(() -> tv.setText("Completed!"));
    }).start();
  }
}
```


Step 3/5
Start Progress


Completed!
Start Progress

7. Write a program to develop an native application that uses GPS location Information.

Activity_main.xml:-

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
   android:orientation="vertical"
   android:gravity="center"
   android:layout_width="match_parent"
   android:layout_height="match_parent"
   android:padding="20dp"
   android:background="#E8F5E9">

   <TextView
      android:id="@+id/textView"
      android:layout_width="wrap_content"
      android:layout_height="wrap_content"
      android:text="Fetching Location..."
      android:textSize="20sp"
      android:textStyle="bold"
      android:textColor="#2E7D32"
      android:gravity="center"
      android:padding="16dp" />

</LinearLayout>
```

MainActivity.java:-

```
package com.example.androidpracticals;

import android.Manifest;
import android.content.pm.PackageManager;
import android.location.Location;
import android.location.LocationListener;
import android.location.LocationManager;
```

```
import android.os.Bundle;
import android.widget.TextView;
import android.widget.Toast;

import androidx.appcompat.app.AppCompatActivity;
import androidx.core.app.ActivityCompat;
import androidx.core.content.ContextCompat;

public class MainActivity extends AppCompatActivity {

  private static final int LOCATION_PERMISSION_REQUEST = 1;
  private LocationManager locationManager;
  private TextView textView;

  @Override
  protected void onCreate(Bundle savedInstanceState) {
    super.onCreate(savedInstanceState);
    setContentView(R.layout.activity_main);

    textView = findViewById(R.id.textView);
    locationManager = (LocationManager) getSystemService(LOCATION_SERVICE);

    // Check and request location permission
    if (ContextCompat.checkSelfPermission(this,
Manifest.permission.ACCESS_FINE_LOCATION)
        != PackageManager.PERMISSION_GRANTED) {
      ActivityCompat.requestPermissions(this,
          new String[]{Manifest.permission.ACCESS_FINE_LOCATION},
LOCATION_PERMISSION_REQUEST);
    } else {
      getLocation();
    }
  }

  // Method to get current location
  private void getLocation() {
    try {
      locationManager.requestLocationUpdates(
          LocationManager.GPS_PROVIDER,
          2000, // Minimum time interval between updates (in ms)
          5,   // Minimum distance between updates (in meters)
```

```
            new LocationListener() {
              @Override
              public void onLocationChanged(Location location) {
                double lat = location.getLatitude();
                double lng = location.getLongitude();
                textView.setText("Latitude: " + lat + "\nLongitude: " + lng);
              }

              @Override
              public void onStatusChanged(String provider, int status, Bundle extras) {}

              @Override
              public void onProviderEnabled(String provider) {
                Toast.makeText(MainActivity.this, "GPS Enabled",
Toast.LENGTH_SHORT).show();
              }

              @Override
              public void onProviderDisabled(String provider) {
                Toast.makeText(MainActivity.this, "Please enable GPS",
Toast.LENGTH_LONG).show();
              }
            });
    } catch (SecurityException e) {
      e.printStackTrace();
    }
  }

  // Handle permission result
  @Override
  public void onRequestPermissionsResult(int requestCode, String[] permissions,
int[] grantResults) {
    super.onRequestPermissionsResult(requestCode, permissions, grantResults);
    if (requestCode == LOCATION_PERMISSION_REQUEST) {
      if (grantResults.length > 0 && grantResults[0] ==
PackageManager.PERMISSION_GRANTED) {
        getLocation();
      } else {
        Toast.makeText(this, "Permission Denied!", Toast.LENGTH_SHORT).show();
      }
    }
```

```
    }
}
```

7:53
Fetching Location...


7:53
Latitude: 37.421998333333335
Longitude: -122.084

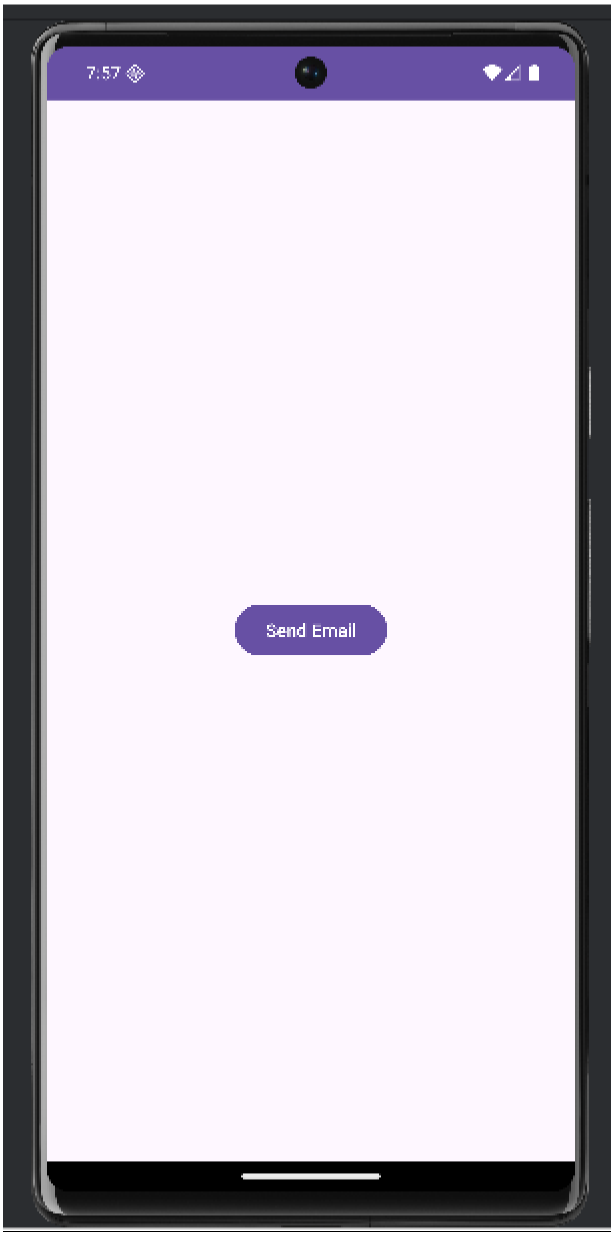
8. Write a program to develop a mobile application to send an E-mail.

Activity_main.xml:-

```
<Button xmlns:android="http://schemas.android.com/apk/res/android"
   android:id="@+id/btnEmail"
   android:text="Send Email"
   android:layout_width="wrap_content"
   android:layout_height="wrap_content"
   android:layout_gravity="center"/>
```

MainActivity.java:-

```
package com.example.androidpracticals;
import androidx.appcompat.app.AppCompatActivity;
import android.os.Bundle;
import android.content.Intent;
import android.net.Uri;
import android.widget.Button;

public class MainActivity extends AppCompatActivity {
   @Override
   protected void onCreate(Bundle s) {
      super.onCreate(s);
      setContentView(R.layout.activity_main);

      Button b = findViewById(R.id.btnEmail);
      b.setOnClickListener(v -> {
         Intent email = new Intent(Intent.ACTION_SENDTO);
         email.setData(Uri.parse("mailto:recipient@example.com"));
         email.putExtra(Intent.EXTRA_SUBJECT, "Demo Subject");
         email.putExtra(Intent.EXTRA_TEXT, "This is a demo email.");
         startActivity(email);
      });
   }
}
```


7:57
Send Email


8:03
From lakshay22saini@gmail.com
To recipient@example.com
Subject